DETECTING EMOTIONAL DISTRESS IN REDDIT POSTS USING NATURAL LANGUAGE PROCESSING

DATA 641 • Applied Natural Language Processing

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Project focus

Investigating whether simple, interpretable NLP models can detect expressions of emotional distress from Reddit post content.

Reddit offers informal, semi-anonymous language where people may describe distress outside clinical or structured settings.

Research question and design choice

The project focuses on post content rather than using subreddit membership as a proxy for emotional state.

Research question

How effectively can traditional NLP models detect emotional distress in individual Reddit posts?

Models tested: TF-IDF features with logistic regression, linear SVM, and a simple neural network.

Key design choice

Manual post-level labels based on content, not subreddit-level labels.

This matters because posts within the same subreddit can vary widely: a mental-health subreddit post may be informational, while a general subreddit post may contain distress-related language.

Data source and sample

Low et al. (2020) Reddit mental health corpus; 400 posts manually labeled for supervised learning.

Labeled sample

400

manually labeled posts

Training and held-out testing used this post-level labeled subset.

Purpose: build and evaluate a classifier, not estimate prevalence.

Mental health-focused subreddits

15 specific support groups
- r/depression
- r/anxiety
- r/suicidewatch
- r/ptsd
- r/addiction
- plus EDAnonymous, alcoholism, ADHD, autism, bipolarreddit, BPD, healthanxiety, lonely, schizophrenia, socialanxiety

2 broad mental health subreddits: r/mentalhealth and r/COVID19_support

General-interest subreddits

11 non-mental-health communities
- r/personalfinance
- r/relationships
- r/legaladvice
- r/parenting
- r/fitness
- r/jokes
- plus conspiracy, divorce, guns, meditation, teaching

These provided baseline and non-distress-oriented content for comparison.

Manual labeling strategy

Manual post-level labels were guided by a coding guide, not subreddit membership or keyword counts.

Label = 1
Distress-related

- Personal emotional strain or difficulty coping
- Sadness, overwhelm, loneliness, exhaustion, or hopelessness
- Direct or indirect expressions of internal distress

Ambiguous cases

Judged using overall tone, context, and personal emotional impact—not keyword presence alone.

Label = 0
Non-distress

- External topics, neutral information, or general discussion
- Complaints or stressful situations without clear emotional impact
- Humor, sarcasm, or practical problem-solving without distress

Intentional coding process

Label definitions + decision checklist clarified inclusion/exclusion criteria and supported consistent decisions.

Modeling pipeline

A deliberately simple and interpretable pipeline: clean text, extract TF-IDF features, compare classifiers.

Raw posts → Preprocess → TF-IDF → Train models → Evaluate

Preprocessing

Tokenization • lowercasing • stop word removal • punctuation and non-text cleanup

Models compared

Logistic Regression
Linear SVM
Feedforward Neural Network / MLP

Evaluation priorities

Because the objective is to identify distress-related posts, the precision–recall tradeoff matters.

Recall

Of the truly distress-related posts, how many did the model catch?

Prioritized because false negatives mean missed distress-related language.

Precision

Of the posts flagged as distress, how many were actually distress-related?

Important because overclassification creates false positives.

F1-score

How well did the model balance precision and recall?

Useful for comparing models when both error types matter.

Accuracy was reported, but the main interpretation depends on sensitivity to distress and false-positive control.

Model comparison

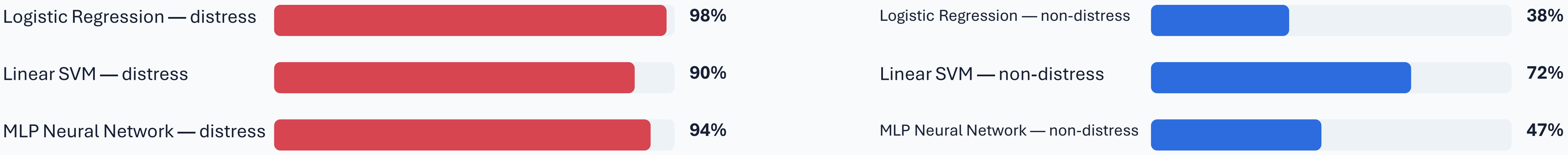
Linear SVM had the best overall balance across classes.

Accuracy

Winner: Linear SVM

Accuracy = 0.825
Distress recall = 0.90
Non-distress recall = 0.72

Recall by class

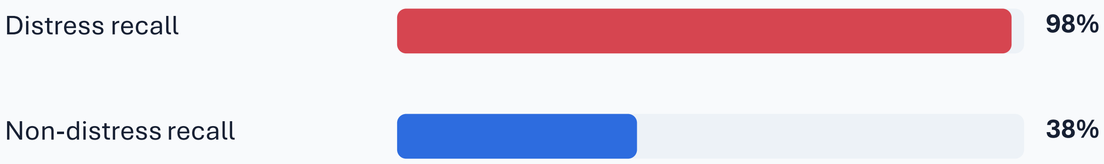


What changed from Logistic Regression to Linear SVM?

The SVM reduced false positives while preserving strong detection of distress.

Logistic Regression

Very sensitive to distress language, but tended to classify many neutral posts as distress.



Linear SVM

Slightly less sensitive for distress, but much better at separating non-distress content.

Interpretation: the SVM gave the best balance between sensitivity and false-positive control.

What language signaled distress?

Distress-related posts combined explicit emotion, internal reflection, coping difficulty, and informal immediacy.

Explicit emotion

depressed | anxiety | hate | suicide

Clear references to emotional suffering, crisis, or distress-related states.

Coping difficulty

struggling | tired | anymore

Language suggested ongoing strain or difficulty continuing.

Internal framing

feel | feeling | life | time

Posts often centered on personal experience, emotional state, and broader life circumstances.

Informal / Urgent expression

im | ve | don | just

Conversational, unstructured language may reflect urgency, spontaneity, or emotional immediacy.

What language signaled non-distress?

Non-distress posts tended to use external, practical, and topic-specific language rather than introspective emotional language.

Practical / informational domains

credit | company | invest | school

Focused on factual discussion, advice-seeking, or practical problem-solving.

Advice or problem-solving

legal | money | relationship | work

Situational concerns discussed in a practical or solution-oriented way.

Everyday life contexts

girlfriend | boyfriend | son | birthday | dating

Life topics often framed externally rather than as emotional distress.

Reddit / structural artifacts

https | subreddit

Some features reflected links or platform structure rather than emotional content.

Caution: some non-distress signals may reflect subreddit topics and dataset structure, not emotional state alone.

Main interpretation

The model appears to separate internal emotional experience from external situation-focused discussion.

Distress posts

Internal
subjective
emotional
immediate

"How I feel / what I can't handle"

Non-distress posts

External
situational
informational
problem-focused

"What happened / what should I do?"

Limitations

The results are promising, but several limitations affect interpretation and generalizability.

Small labeled dataset

The broader corpus was larger, but supervised models used 400 manually labeled posts. This limits model complexity and generalizability.

Sampling bias

Structured sampling ensured distress examples, but some class differences may reflect topic or subreddit composition.

Manual annotation subjectivity

Labels were applied consistently, but future work should use multiple annotators and inter-rater reliability.

Post text only

The analysis did not include user history, comment responses, or other contextual information.

These limitations point to the need for larger labeled samples, stronger annotation procedures, and broader validation.

Takeaways and next steps

Interpretable NLP methods can identify meaningful patterns in how distress is expressed online.

Best model: Linear SVM

Accuracy 0.825 • Distress precision 0.83 • Distress recall 0.90 • F1-score 0.86

Takeaways

1. Post-level labels avoided relying on subreddit membership as the outcome.
2. TF-IDF with linear models performed well in a small labeled dataset.
3. Feature analysis showed distress language was emotionally expressive and introspective.

Future work

Expand the labeled dataset
Use semi-supervised or weak supervision approaches
Add context from comment responses or user history
Compare with more advanced language models

Thank you